

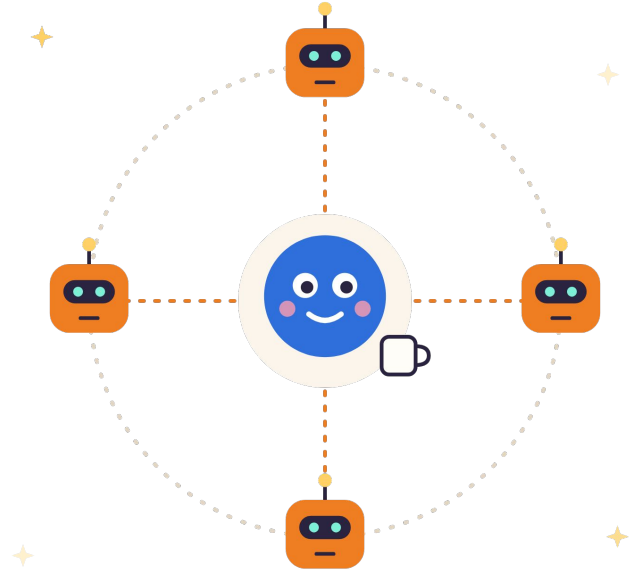
Adapting Socio-Technical Congruence to Human-Agent Collectives

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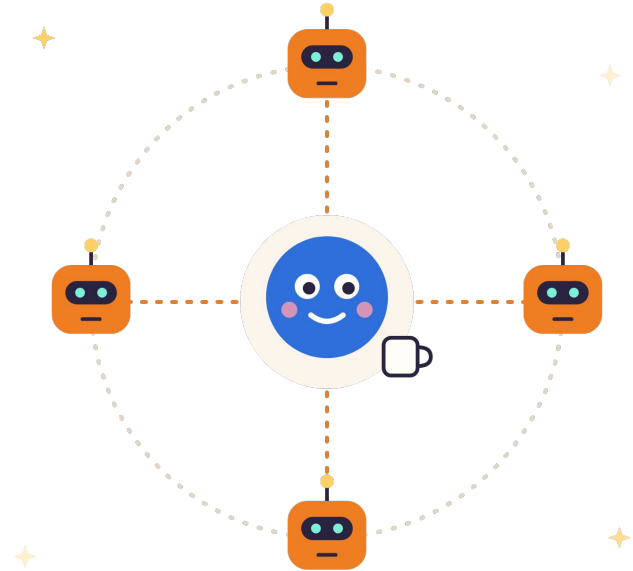
Sept. 28, 2026



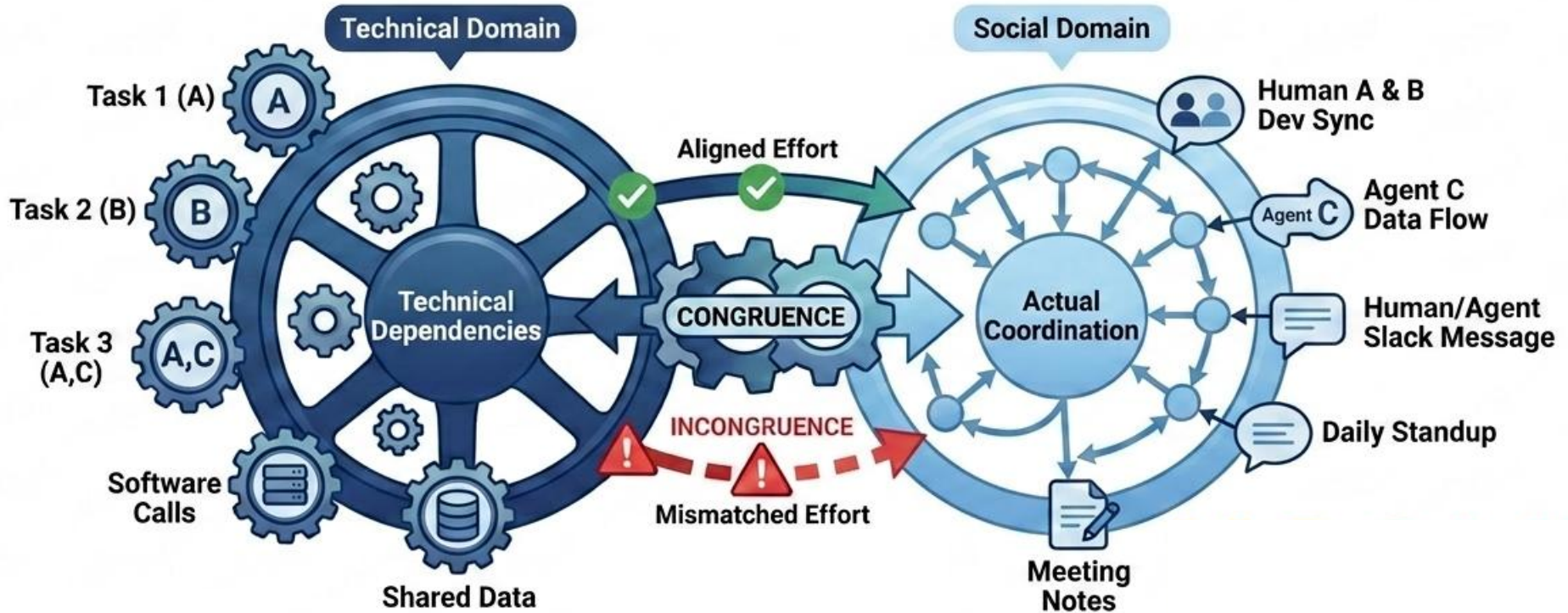
Who uses agents daily?



Who uses agents for software dev?



Socio-Technical Congruence



Socio-Technical Congruence (Cataldo et al.)

How well the communication of developers aligns with the dependencies and technical constraints in their code

$$CR = (TA \cdot TD \cdot TA^T)$$

CR: coordination-requirements matrix
(who should be coordinating?)

TD: task dependencies
(which work depends on other work)

TA: task assignments
(who works on what?)

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$$\text{Congruence}(CR, CA) = \frac{\text{Diff}(CR, CA)}{|CR|}$$

Socio-Technical Congruence (Cataldo et al.)

- Incongruence means friction
- Congruent tasks get done faster

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(who should be coordinating?)

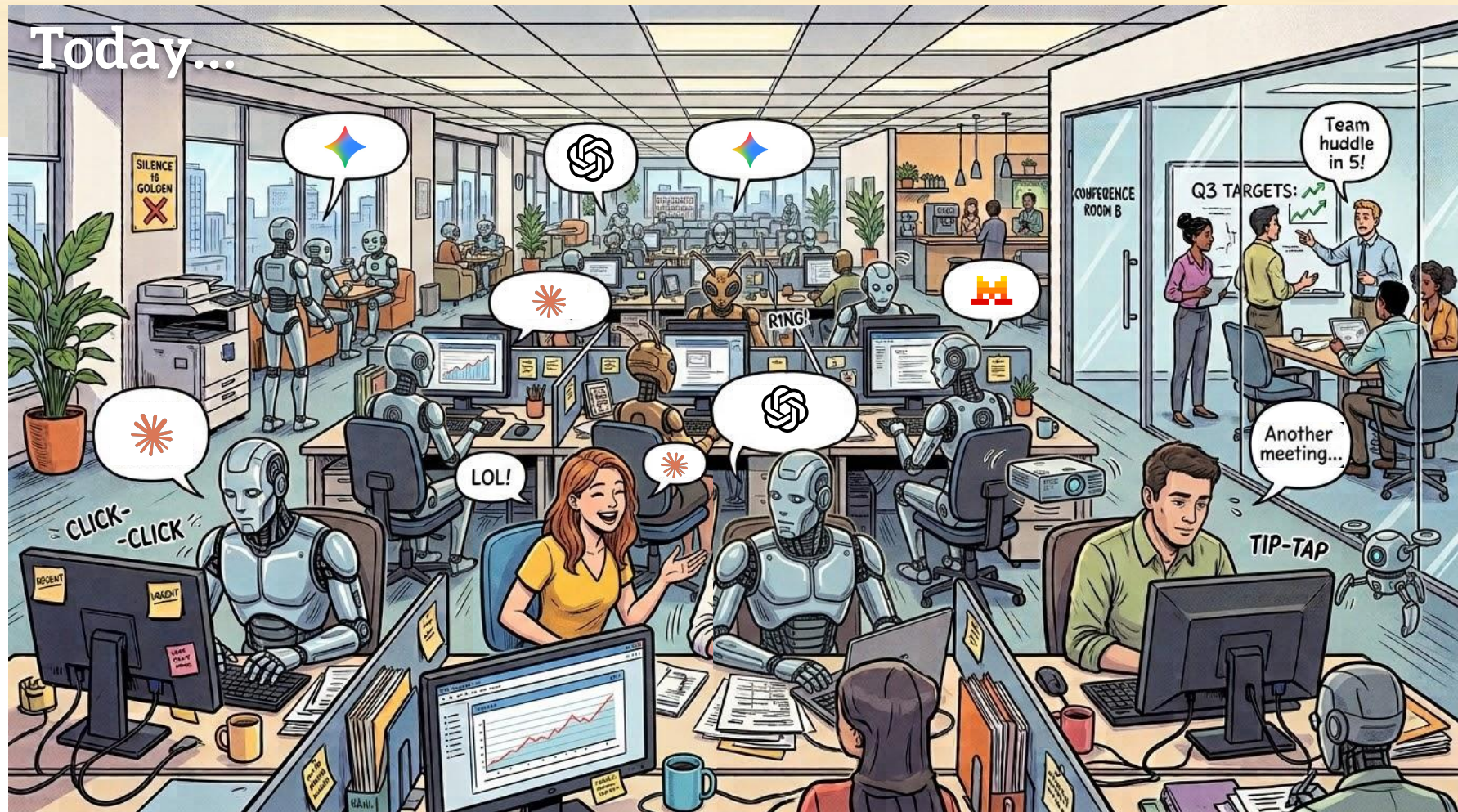
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Today...



Scope Conditions of the Original Construct



Homogeneous nodes

Everyone replies at similar speed, remembers for days, and can be watched working.

Human workers

- cognitive capacity
- speed (9-to-5'ing)
- observability

Social Domain

Scope Conditions of the Original Construct



Static assignments

Who works on what is one snapshot for the whole task.

Technical Domain

Task Assignments (TA)
are static.

Task Dependencies (TD)
never change.

Scope Conditions of the Original Construct



Observable channels

Coordination leaves a trace: emails, commits, change requests.

Technical Domain

Coordination activities (CA)
are recovered from observable
artifacts (commits, emails, logs, ...)

Scope Conditions of the Original Construct

</> A4

Code-level ties

Dependencies come from the code: call graphs, builds, shared files.

Technical Domain

Task Dependencies (TD)
are recovered from the code
from call graphs, build-time
references, shared file edits

Scope Conditions of the Original Construct



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Static assignments

Who works on what is one snapshot for the whole task.



Observable channels

Coordination leaves a trace: emails, commits, change requests.



Code-level ties

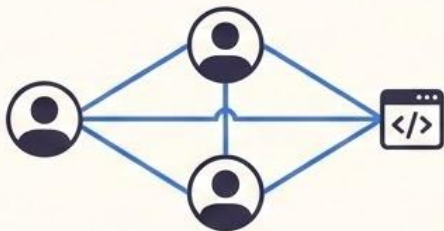
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Adapting Sociotechnical Congruence to Human-Agent Collectives

Sociotechnical Congruence requires revision for modern systems

Designed for purely human software teams, its core assumptions break down in hybrid collectives.

Classic Context



- Observable human actors
- Static assignments
- Code-level ties

Hybrid Context (Humans + Agents)



- Mismatched latency (hours vs. seconds)
- Hidden interaction traces
- Latent context coupling (shared training/tools)

Adapting Sociotechnical Congruence to Human-Agent Collectives



hours

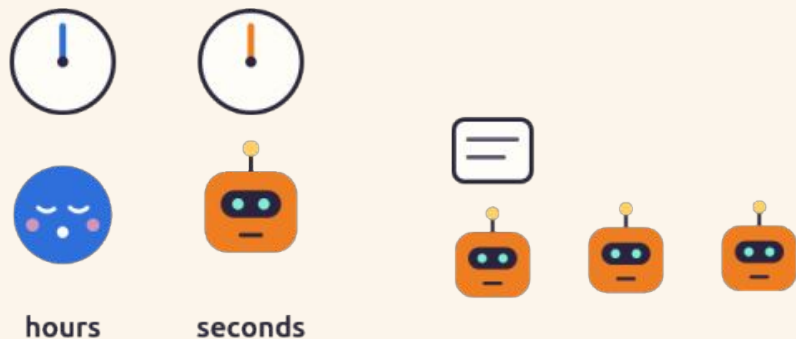


seconds

A1: Mismatched clocks

Agents act in seconds; humans answer in hours. Memory and visibility differ too.

Adapting Sociotechnical Congruence to Human-Agent Collectives



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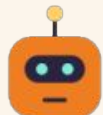
A2: Work reshuffles

Orchestrators reassign tasks at machine speed; agents spawn unassigned helpers.

Adapting Sociotechnical Congruence to Human-Agent Collectives



hours



seconds



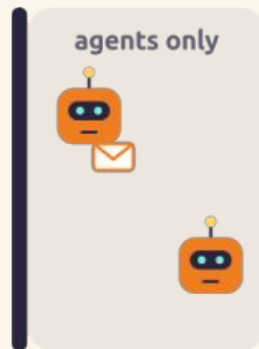
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A3: Hidden communication

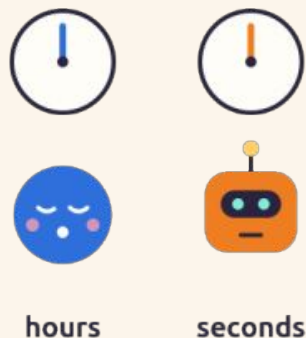
Agent-to-agent handoffs live in tool logs that humans rarely read.



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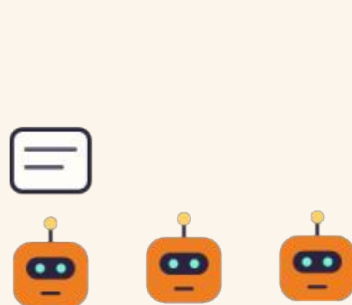
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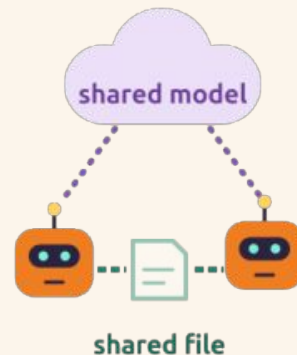
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A4: Invisible ties

Shared context, tools and training couple agents without any call-graph edge.

Typology of Hybrid (In)Congruences

Latency

breaks A1

Clocks too far apart: the need changes before the slow one replies.

Awareness

breaks A3

Handoffs move to channels no human watches.

Memory

breaks A1

Agents forget; losses pile up along long chains.

Self-initiation

breaks A2

An agent joins the work without being assigned.

Assignment churn

breaks A2

Work hops between agents faster than anyone tracks.

Context-coupling

breaks A4

Agents collide through a shared file or memory store.

Training

breaks A4

Same base model, same mistakes, logged as independent.

Accountability

breaks A1 + A3

Nobody can reconstruct who validated what.

A1: Mismatched clocks

A2: Work reshuffles

A3: Hidden communication

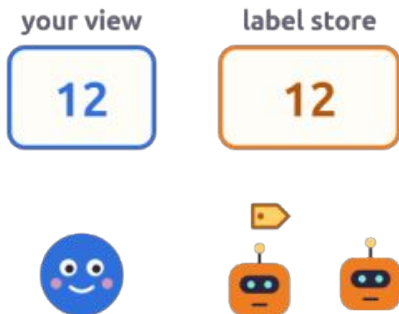
A4: Invisible ties

Examples of Breakdowns (1/2)



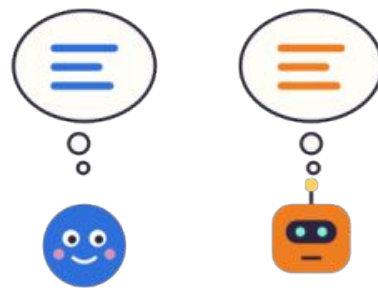
Latency

Agents continue to work over night. Human reviews the patch in the morning. Activities not traced in CA.



Awareness

Agents keep revising a shared label store. Human has a stale mental model and works from an outdated state.



session 1

Memory

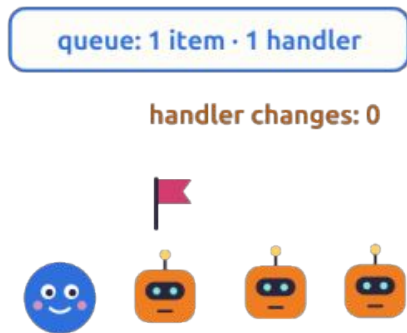
The agent's context resets between sessions. The user still remembers.



Self-initiation

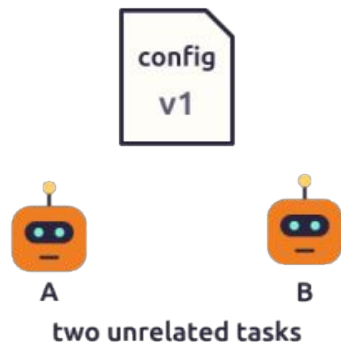
A reviewer agent spawns a helper to fix a bug outside of its original task. → Change in Task Assignments (TA)

Examples of Breakdowns (2/2)



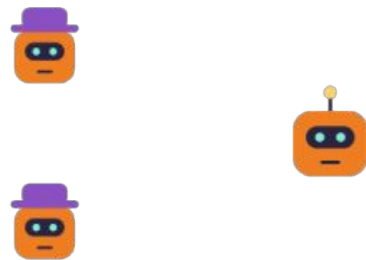
Assignment churn

Agents reassign work among sub-agents.
The human reviewer's queue shows one item.



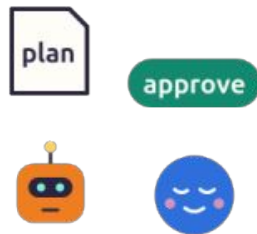
Context-coupling

Agent A rewrites a config as a side effect.
Agent B quietly adopts it.



Training

Same-vendor agents make the same mistake, logged as independent sources.



Accountability

An agent proposes an experiment. The human approves unread.
Recorded in CA, but without justifications.

Hybrid Congruence

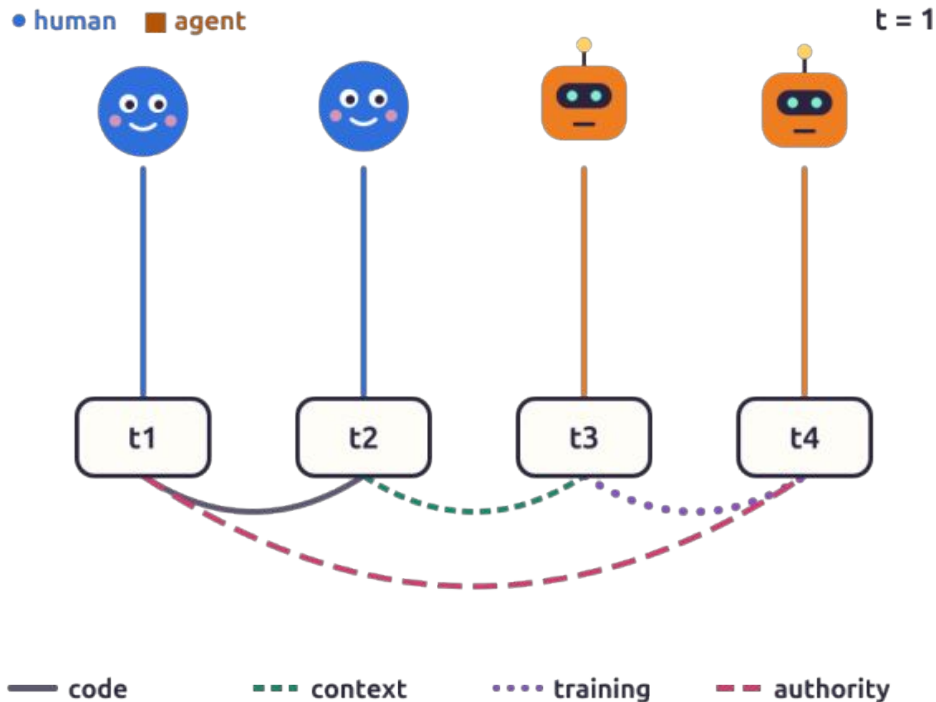
$$CR^*(t) = TA(t) \cdot TD^* \cdot TA(t)^T$$

Typed actors. Every row of TA is labeled human or agent.

Time-varying assignments. TA(t) can change, and even grow, mid-task.

Four edge types in TD*. Code, context-coupling, training, authority.

Congruence is then scored within each type and at the human-agent boundary, not as one number.



Parameterizing the Typology: Simulation (Toy) Model

1 + 4

one human coordinating four
agents

200

Monte Carlo trials per parameter
value

95% / 30%

coordination state kept per step,
human vs. agent

12

tasks in each random dependency
graph

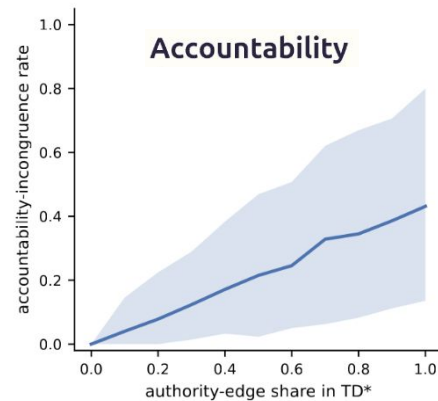
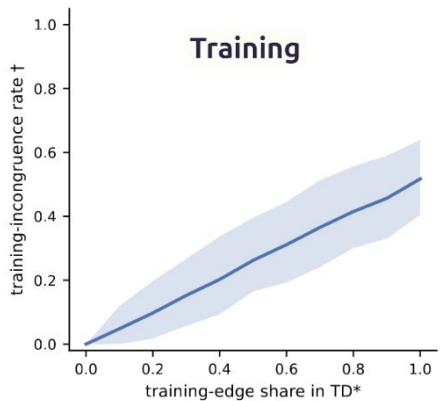
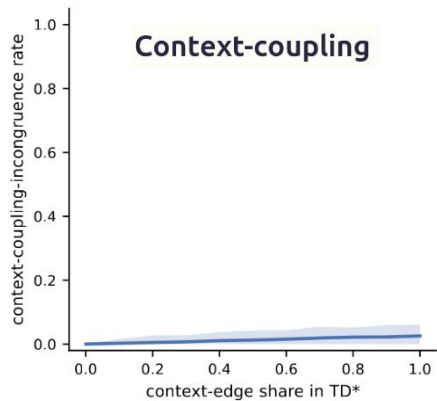
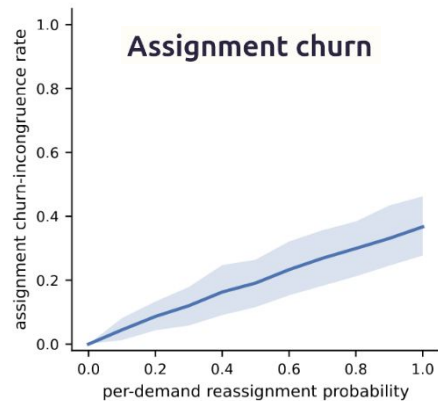
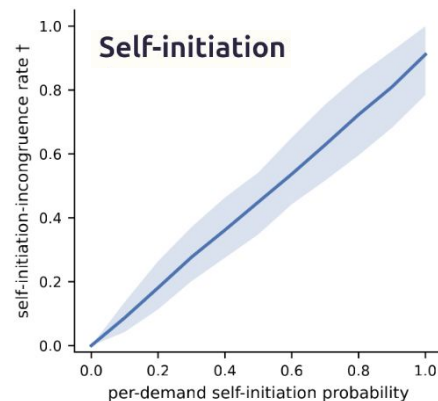
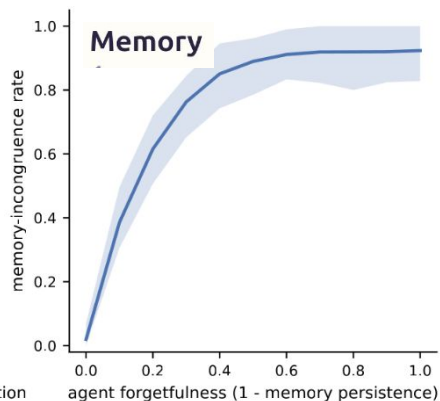
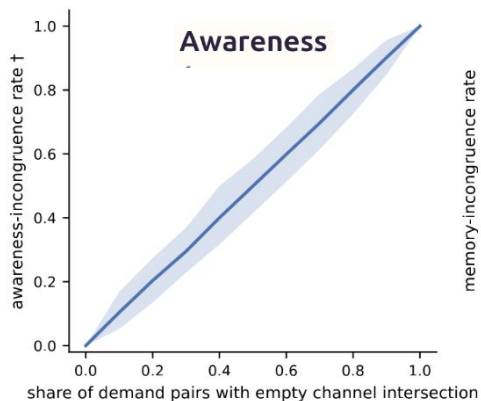
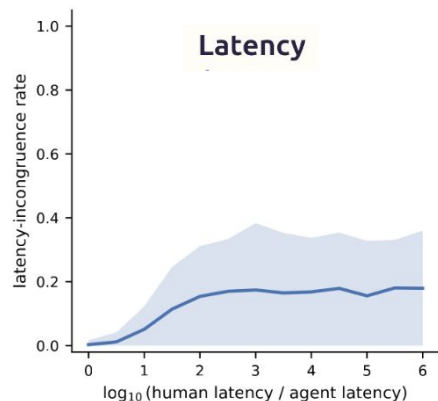
4 h / 20 s

typical human vs. agent reply time

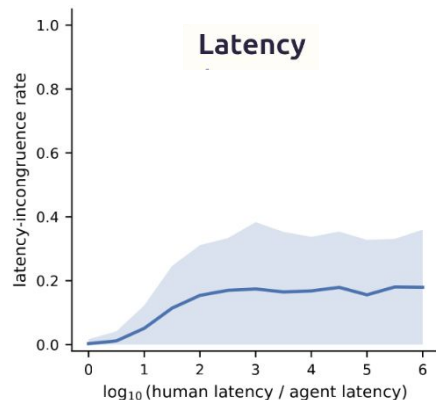
Sweeps

1 parameter at a time; the rest held fixed

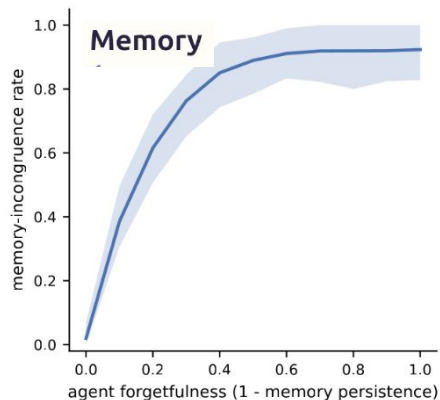
Simulation Model Results



Simulation Model Results



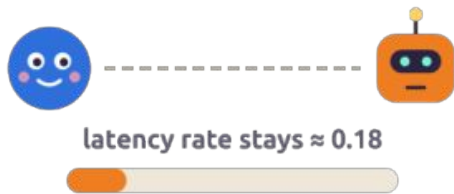
The curve rises and then flattens, once the agent is about 100x faster than the human. At this point, the human reviews one in five agent tasks and this sets the ceiling.



The curve climbs and levels off. This is caused by human forgetting. Even moderate forgetting in human disrupts most coordination demands that reach an agent.

Predictions

human 100× slower



Speed isn't the lever

Beyond about 100x speedup of agents, the human–agent latency gap does no longer matter. The limit is how much work is passed between humans and agents.

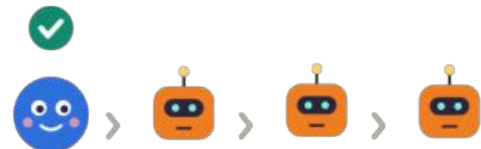
coordination state, hop by hop



Forgetting compounds

Agent memory matters in long dependency chains: every step is another chance to lose the thread.

who validated this?



Blame evaporates

Responsibility is hardest to reconstruct along delegation chains with no human in the loop. Human coverage on approval edges bounds this breakdown.

Design Implications

Mode	Design response
Latency	Restructure which dependencies cross the human-agent boundary
Awareness	Surface the inter-agent log in the human's working view
Memory	Persistent stores and summaries, worth it in proportion to chain depth
Self-initiation	Register spawned sub-agents in the human-visible assignments
Assignment churn	Carry handoff state explicitly across every reassignment
Context-coupling	Give each agent its own namespace in shared environments
Training	Mix vendors in a pipeline: same-vendor agents err together
Accountability	Put human coverage on approval (authority) edges

Limitations

- **All curves and values are illustrative.**
Three of eight curves are linear by construction.
- Several ceilings echo chosen constants,
e.g., 10% handover loss, ratio of 1:4 agents, ...
- Chain depth is proxied by in-degree, capped at 6.
Retry loops are not modeled.
- No estimator for real systems yet. Every curve **needs empirical measurements and validation.**

Open Questions

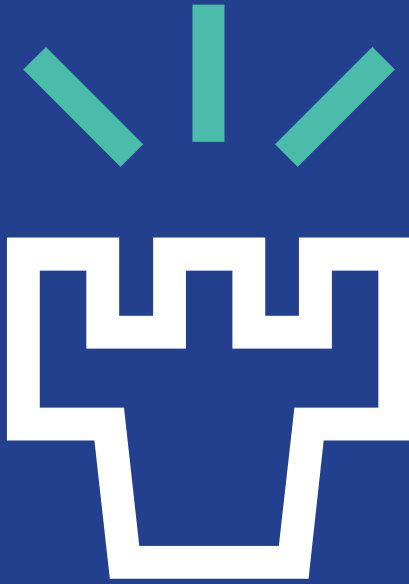
- Should we model hybrid actors?
Is an agent a teammate, a tool, or something in between?
- New types of failures / coordination / communication,
or old ones on steroids?
- Instrumentation harness:
What would we need to log and measure in
human-agent collectives, and how?

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